JAVA TRAINING PROGRAM

Java Loop Programs

1. Print Numbers from 1 to 50

Displays integers from 1 through 50 using a for loop.

```
public class
onetofifty
 {
 public static void
main(String[] args)
{
for (int i = 1; i <=
50; i++)
{
System.out.println(i
);
        }
    }
}
```

2. Print Odd Numbers from 1 to 100

Starts at 1 and increases by 2 to display odd numbers up to 100.

```
public class
onetohundred
{
 public static void
main(String[] args)
{
       for (int i =
1; i <= 100; i += 2)
{
System.out.println(i
);
          }
     }
}
```

3. Print Numbers in Descending Order (100 to 1)

Counts down from 100 to 1 by decrementing i.

```
public class down {
public static void
main(String[] args)
{          for (int i
= 100; i >= 1; i--)
{
System.out.println(i
);
          }
     }
}
```

4. Sum of Numbers from 1 to 100

Adds each value from 0 through 100. The result is 5050.

```
public class
sumnumbers {
public static void
main(String[] args)
{          int sum =
0;          for (int
i = 0; i <= 100;
i++) {
sum = sum + i;
}

System.out.println("
sum=" + sum);
    }
}
```

5. Count Numbers Divisible by 7

Uses the remainder operator (%) to count multiples of 7 from 1 to 100. The result is 14.

```
public class
divisible {
public static void
main(String[] args)
{          int count
= 0;          for
(int i = 1; i <=
100; i++) {
if (i % 7 == 0) {
count++;
}
         }
         System.out.println("count=" +
count);
     }
}
```

6. Sum of Even Numbers from 2 to 10

Adds 2 + 4 + 6 + 8 + 10. The result is 30.

```
public class
evennumbers {
    public static
void main(String[]
args) {          int
n = 10;          int
sum = 0;          for
(int i = 2; i <= n;
i += 2) {
sum = sum + i;
}
        System.out.println("Sum of even
numbers=" + sum);
    }
}
```

7. Factorial of a Number

Reads a number and calculates its factorial by

multiplying the integers from 1 to n. For example, 5! =

120. `import java.util.Scanner;`

```
public class
factorial
{
    public static
voidmain(String[]
args)
{        Scanner sc = new
Scanner(System.in);
System.out.print("Enter a number: ");
int n = sc.nextInt();
          long fact
= 1;           for
(int i = 1; i <=
n; i++) {
fact = fact * i;
}


System.out.println("Num
ber = " + n);
System.out.println("Fat
```

```
orial = " + fact);
sc.close();     }
}
```

8. Multiplication Table

Reads a number and prints its multiplication table from 1 to 10.

```
import
java.util.Scanner;

public class
multiplication {
public static void
main(String[] args) {
Scanner sc = new
Scanner(System.in);
System.out.print("Ent
er a number: ");
int n = sc.nextInt();

        for (int i = 1; i <=
10; i++) {
System.out.println(n + " x " +
i + " = " + (n * i));
}         sc.close();     }
}
```